

Master thesis

Master of Science (M.Sc.)
Department of Business
Major: Data Science

Update the
title page
with related
text

Topic:

**Enhancing Solar-Induced Fluorescence OCO-2 data
for High-Resolution Crop Yield Prediction in
Germany**

Author: VT

Matriculation Number:
First supervisor: Prof. Dr.
Second supervisor: Mr.
Submitted on: 20.08.2026

Statutory Declaration:

I hereby declare that I have developed and written the enclosed Master Thesis completely by myself and have not used sources or means without declaration in the text. I clearly marked and separately listed all the literature and all the other sources which I employed when producing this academic work, either literally or in content. I am aware that the violation of this regulation will lead to failure of the thesis.

Berlin, xx.xx.202x

.....

(Signature)

[your name]

Abstract

Four sentences of background and importance of this topic.

Four sentences defining the gap analysis (work not done in field) and your problem statement.

Four sentences about the methodology of your work.

Six sentences of Results and significance of your results.

Two sentence summarizing your contribution.

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write a one page abstract
- Approximately 20 sentences with importance, background, current state, gap analysis, problem statement, your solution, your methods, your results, and improvement in field being discussed.

Zusammenfassung

Da die meisten Leuten an der TU deutsch als Muttersprache haben, empfiehlt es sich, das Abstract zusätzlich auch in deutsch zu schreiben. Man kann es auch nur auf deutsch schreiben und anschließend einem Englisch-Muttersprachler zur Übersetzung geben.

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1 Introduction

One-Two paragraph(s) introducing the purpose and theme of the research.

1.1 Motivation

Two-Four paragraph(s).

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1.2 Objective

One-Two paragraph(s).

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'Figure 1.1 shows Component X as part of ...'

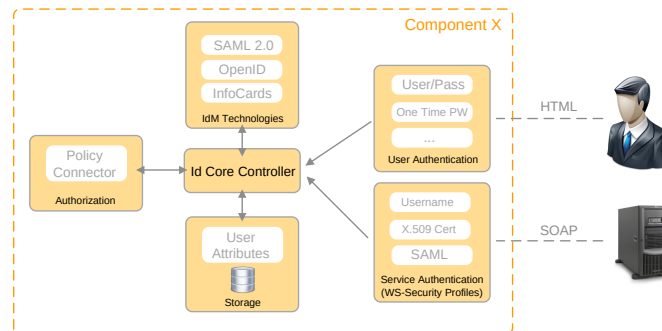


Figure 1.1: Component X

1.3 Thesis Scope

One-Two paragraph(s).

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1.4 Purpose and goal

One-Two paragraph(s).

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1.5 Outline

The 'structure' or 'outline' section gives a brief introduction into the main chapters of your work. Write 2-5 lines about each chapter. Usually master thesis is separated into 6-8 main chapters.

This example thesis is separated into 7 chapters.

Chapter 2 analyzes the requirements for your component. This chapter provides the reader the background knowledge to understand technical concepts applied in the thesis. This chapter will have 5-10 pages.

Chapter 3 is usually termed 'Related Work', 'State of the Art' or 'Fundamentals'. Here you will describe relevant technologies and standards related to your topic. What did other scientists propose regarding your topic? This chapter makes about 20-30 percent of the complete thesis.

Chapter 4 is termed 'Research question'. Here you describe your Problem statement and Research questions.

Chapter 5 describes the methodology/implementation part of your work. Don't explain every code detail but emphasize important aspects of your implementation. This chapter will have a volume of 20-30 percent of your thesis.

Chapter 6 is Experiments, Results and Discussion. This chapter will have a volume of 15-20 percent of your thesis.

Chapter 7 summarizes the thesis, describes the problems that occurred and gives an outlook about future work. Should have about 4-6 pages.

2 Theoretical Background

This chapter provides the reader the background knowledge to understand technical concepts applied in the thesis.

2.1 Background

2.1.1 Definitions

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2.2 Existing Methods

At least 8-10 different cited papers for techniques you deploy for problem solution with 6-8 published after 2019.

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2.3 Evaluation Metrics

2.3.1 Subsection 1

2.3.2 Subsection 2

2.3.3 Subsection 3

3 Related work

In this chapter, we discuss the relevant literature review for the chosen topic.

Identify the important subcomponents (e.g., in terms of methods, evaluation, preprocessing or visualization) of your thesis and for each subcomponent, make a separate section with subsections as shown below.

The summary of the literature is shown in Table ??.

3.1 Section 1

At least 8-10 different cited papers for techniques you deploy for problem solution with 6-8 published after 2023.

Group in terms of similarity in approach.

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3.2 Section 2

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Group in terms of similarity in approach.

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4 Research question

This chapter introduces the research question ...

4.1 Problem statement

4.2 Research question

5 Methods

5.1 Research Design and Study Area

Introduce the overall workflow, Germany/Bavaria study areas, spatial scales, study years, and relationship between SIF enhancement and yield modelling.

5.2 Datasets

Describe OCO-2 SIF, Sentinel-2 composites, MODIS/VIIRS products, crop-type rasters, agro-climatic zones, administrative boundaries, and yield statistics.

5.3 Data Preprocessing

Cover SIF target construction, quality and uncertainty filtering, footprint geometries, temporal matching, raster alignment, masking, and missing-data treatment.

5.3.1 Input Data Representations

Explain polygon-mean tables, single-footprint chips, multi-footprint chips, 4 km spatial aggregation, native-footprint inference samples, and crop-pure yield predictors.

5.4 Model Architectures

Describe Random Forest, XGBoost, the area-to-point MLP, CNN/U-Net architectures, footprint-weighted supervision, and the yield model.

5.5 Model Training and Experimental Design

Present data splitting, leakage prevention, normalization, loss functions, optimization, calibration, external-tile testing, and the 2019–2022/2024 yield split.

5.6 Experimental Environment and Reproducibility

Document software, hardware, Kaggle/local environments, random seeds, checkpoints, and saved model-ready datasets.

5.7 Techniques for Visualization

5.8 Evaluation Metrics

Define RMSE, MAE, bias, R^2 , regression slope, residual analysis, permutation importance, subgroup evaluation, and bootstrap uncertainty.

6 Experiments, Results and Discussions

6.1 Experimental Overview

Briefly introduce the evaluated model configurations, training behaviour, test datasets, and distinction between aggregate, native-footprint, and external evaluation.

6.2 Overall SIF Prediction Performance

Present the main internal and external prediction results, observed-versus-predicted plots, residual behaviour, spatial residual map, and comparison with published enhanced SIF products. This section establishes how accurately enhanced SIF was predicted and how well the 4 km model transferred to native footprints.

6.3 Effects of Model Architecture and Input Representation

Compare Random Forest, XGBoost, MLP, single-footprint CNN, multi-footprint CNN, and spatial-aggregate U-Net results. Discuss tabular versus spatial inputs, example SIF maps, residual distributions, and feature importance.

6.4 Spatial Aggregation, Stability and Generalization

Compare native-footprint and 4 km supervision, including target variability, residual stability, footprint-count effects, and the trade-off between retrieval-noise reduction and spatial detail. External testing on unseen MGRS tiles can also be interpreted here.

6.5 Spatial and Seasonal Variation in Model Performance

Present performance across agro-climatic zones, months, measurement modes, years, and geographic regions. Include the zone choropleths, residual boxplots, and zone-by-month heatmap.

6.6 Winter-Wheat Yield Prediction

Compare crop-pure enhanced SIF with raw mixed-pixel OCO-2 SIF for yield prediction. Present model performance, observed-versus-predicted yield, paired regional errors, and bootstrap RMSE improvement.

6.7 Comparison with Existing Techniques and Integrated Discussion

Bring the findings together, compare them with related studies, and discuss Benefits of spatial aggregation Value of high-resolution predictor structure Limitations of native-footprint

transfer Agro-climatic variability Usefulness of crop-pure SIF for yield modelling Method-
ological limitations and practical applicability

7 Conclusion

The final chapter summarizes the thesis. The first subsection outlines the main ideas behind Component X and recapitulates the work steps. Issues that remained unsolved are then described. Finally the potential of the proposed solution and future work is surveyed in an outlook.

7.1 Summary

Explain what you did during the last 6 month on 2 or 3 pages!

The work done can be summarized into the following work steps

- Analysis of available technologies
- Selection of 3 relevant services for implementation
- Design and implementation of X on Windows
- Design and implementation of X on mobile devices
- Documentation based on X
- Evaluation of the proposed solution

7.2 Dissemination

Who uses your component or who will use it? Industry projects, EU projects, open source...? Is it integrated into a larger environment? Did you publish any papers?

7.3 Problems Encountered

Summarize the main problems. How did you solve them? Why didn't you solve them?

7.4 Outlook

Future work will enhance Component X with new services and features that can be used ...

Bibliography

Annex

```
<?xml version="1.0" encoding="UTF-8"?>
<widget>
    dummy source code
</widget>
```

Listing 1: Sourcecode Listing

dummy text

Listing 2: SIP request and response packet[?]